Unit Test Automation Summary Report

Flask Public Blogging Site | Assignment 3

This report summarizes the group project work against the Assignment 3 requirements. The team selected and configured a Python Flask application, developed tests for six application modules across two members, documented test cases, and configured pytest suites for grouped and parallel execution.

Project overview

The selected open-source project is a Flask public blogging application. The local development environment was prepared so the application and its tests could be run. The assignment also calls for a short setup video to accompany the repository submission.

Group members and module allocation

Group member	Application modules selected	Test run
Umar Munir	app/forms.py; app/markdown_utils.py; app/models.py	65 passed
Ali Qasim	app/routes.py; app/auth/routes.py; app/__init__.py	64 passed

Task 2 | Automated unit test cases

Each member developed a suite for three selected modules. Test design notes use equivalence class partitioning and boundary value analysis, with positive and negative cases covering form validation, Markdown rendering and sanitization, data models, authentication, route access control, and application creation. Tests and documentation are organized in separate member folders for individual contribution.

Task 3 | Test case documentation and external data

Each member's workbook documents 10 test cases, for 20 documented cases total. Umar's registration test suite reads seven data rows from a CSV file. The rows exercise valid and invalid username and password boundaries, confirmation matching, and maximum username length. The test data is consumed through parameterized tests.

Task 4 and execution results

The pytest setup separates tests into member and functional-area suites using markers. It supports parallel execution with pytest-xdist, parameterized tests where applicable, and deterministic serial ordering. Each member folder includes configuration and run instructions.

Suite	Passed	Failed
Umar Munir	65	0
Ali Qasim	64	0
Combined run	129	0

All 129 automated tests passed in the reported local runs, with no test failures.

Prepared project materials

Separate member test folders; pytest configuration and run instructions; one test design document and Excel test case workbook per member; and a CSV registration data source for Umar's parameterized tests.